

Boolean laws

Algebra lets you simplify logic before you draw gates

Rewrite with a reason

- Each click should name the law: not of A and B becomes not-A or not-B
- A plus A-and-B collapses to A
- A and not-A is zero
- Double negation peels to A
- Factor A-and-B plus A-and-C into A-and the sum of B and C
- The history list shows your proof chain; Undo when a step goes wrong

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Boolean law playground

Rewrite expressions one law at a time — De Morgan, absorption, complements, and more. Starter walks $\sim(A \cdot B)$ to $A' + B'$. Challenges mix identify-the-law and pick-the-rewrite.

Starter example: $\sim(A \cdot B)$ — apply De Morgan to get $A' + B'$.

Challenge 0 / 22 cleared

Quiz: De Morgan AND: $\sim(A \cdot B)$ equals...

☐ $A' + B'$

☐ $A' \cdot B'$

☐ $A \cdot B$

☐ $A + B$

Show hint

Check

Next

Load into playground

Idle

Quiz: De Morgan AND

Quiz: De Morgan OR

Quiz: double negation

Quiz: complement

Quiz: null element

Quiz: absorption

Quiz: idempotent

Quiz: distribute

Pick: De Morgan step

Pick: absorption

Pick: complement AND

Pick: factor

Pick: identity

Pick: De Morgan OR

Reach: De Morgan

Reach: absorb

Reach: $\sim\sim A$

Reach: $A + A'$

Reach: factor form

Name the law

Name: absorption

Name: complement

Boolean laws starter

Workbook practice

- In the workbook track, simplify not of A or B by De Morgan
- Collapse A plus A-and-B to A
- Show A-and-not-A is zero and A-or-not-A is one
- Factor A-B plus A-C by hand
- Name one pitfall: distributing a NOT only over part of a term

Pitfalls to watch

- Do not treat AND and OR as ordinary arithmetic, De Morgan flips both operator and negation
- Skipping steps can hide an illegal rewrite
- And remember: the browser lab is literacy
- Real synthesis still needs consistent don't-care and timing context

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, do two rewrites with the law named beside each step
- When you are ready, take the short quiz, then continue to K-maps